167181 - PART A

Calculations in series

a) Total resistance in ohms
= 20 + 30 + 50 = 100ohms

b) Total Current form the power supply
V = 1R
125 = I100
I =125/100
1=1.25A

c) Calculate the current through each resistor
In a series circuit, current is the same through all the resistors hence 1.25A

d) Calculate the voltage drop in each resistor
In 20ohms, (V=IR) 20*1.25 =25V
In 30ohms, (V=1R) 30*1.25=37.5V
In 50ohms, (V=IR) 50*1.25=62.5

e) Calculate the power dissipated through each resistor (Power =VI)
In 20ohms, (P=VI) 25*1.25 =31.25w
In 30ohms, (P=V1) 37.5*1.25=46.875w
In 50ohms, (P=VI) 62.5*1.25=78.125w

Calculations in parallel

f) Total resistance in ohms
= 1/Total Resistance = 1/20 + 1/100 + 1/50 =0.08
1/Total resistance = 8/100
Total resistance = 100/8 =12.5
=12.5 ohms

g) Total Current form the power supply
V = 1R
125 = I12.5
I =125/12.5
1=10A

h) Calculate the current through each resistor
In a parallel circuit, voltage is the same through all the resistors hence 125V
V=1R
In 20ohms, (125=20I) 125/20=6.25A
In 100ohms, (125=100I) 125/100=1.25A
In 50ohms, (125=50I) 125/50=2.5A

i) Calculate the voltage drop in each resistor
In a parallel circuit, Voltage is the same in each resistor
hence= 125V

j) Calculate the power dissipated through each resistor (Power =VI)
In 20ohms, (P=VI) 125*6.25=781.25w
In 100ohms, (P=V1) 125*1.25=156.25w
In 50ohms, (P=VI) 125*2.5=312.5w

Part B

Component Table list in Series

Component Reference Name	**Component**	**Value**	**Quantity**
V	DC Voltage source	125v	1
R1	Resistor	20	1
R2	Resistor	30	1
R3	Resistor	50	1
-	Connecting wires	-	4
-	Ground	-	1

Component Table list in Parallel

Component Reference Name	**Component**	**Value**	**Quantity**
V	DC Voltage source	125v	1
R4	Resistor	20	1
R5	Resistor	100	1
R6	Resistor	50	1
-	Connecting wires	-	6
-	Ground	-	1

Component table for DHT22 and Arduino

Component Reference Name	Component	Value	Quantity
Arduino Nano V3	Microcontroller	-	1
DHT22	Temperature and Humidity sensor	-	1
	Connecting wires	-	4